

2.3) Euclidean Distance Modelling for Node Coverage

Integer Linear Programming (ILP) Formulation: The final node placement is solved using the `intlinprog` algorithm, which optimizes the deployment based on the following constraints:

2.3.1) Objective Function (Cost Minimization), given by:

$$\min Z = \sum_{j=1}^N x_j, \quad x_j \in \{0, 1\} \quad (\text{eq.16})$$

Where x_j is a binary decision variable indicating whether a node is deployed (**1**) or not (**0**), Z is the total number of deployed nodes.

2.3.2) Coverage Constraint (Set Cover Problem): Ensures that every pixel in the D matrix is covered by at least one active node, given by:

$$\sum_{j=1}^N A_{i,j} x_j \geq 1, \quad \forall i \in \text{Dead Zones} \quad (\text{eq.17})$$

3. Node Optimization

This technical specification details the mathematical framework designed to optimize the deployment and spatial configuration of Wi-Fi nodes. The model executes a multi-stage pipeline, transitioning from raw visual data to a refined spatial topology that maximizes coverage while strictly mitigating co-channel interference.

3.1) Image Pre-processing: Gray-Level Transformation

The optimization begins with the dimensionality reduction of the input signal. The raw image data, represented as a three-dimensional array $f(x, y, \mathbf{3})$, must be converted into a single-dimension grayscale array $g(x, y)$. This step is critical to minimize the computational overhead required for the subsequent Circular Hough Transform (CHT).

To quantify this transformation, we apply the "Perceptual Weighting" formula to the Red (R), Green (G), and Blue (B) channels:

$$g(x, y) = wr \cdot R(x, y) + wg \cdot G(x, y) + wb \cdot B(x, y) \quad (\text{eq.18})$$

Technical Specification: ITU-R BT.601 Standard To ensure digital standardization and luminance accuracy, the model utilizes the following weighting coefficients:

wr (Red): 0.2989

wg (Green): 0.5870

wb (Blue): 0.1140

Executing this transformation yields the finalized grayscale equation:

$$g(x, y) = 0.2989R + 0.5870G + 0.1140B \quad (\text{eq.19})$$

3.2) Node Detection via Circular Hough Transform (CHT)

The model identifies potential Wi-Fi node proxies—specifically targeting "**black point centers**" within the image space—using the Circular Hough Transform. This methodology maps visual features into a "Parameter Space" to isolate circular geometries.

The parametric equations for the candidate circles are defined as:

$$x = a + r \cos(\theta) \quad (\text{eq.20})$$

$$y = b + r \sin(\theta) \quad (\text{eq.21})$$

Accumulation Logic

The algorithm performs an accumulation process within a 3D matrix $H(a, b, r)$. Within this **3D** accumulator, the Local Maxima identify the most probable (a, b) coordinates for a specified radius r . These maxima correspond to the centers of the visual markers identified in the source image.

Initial Extraction

To extract the initial set of potential node coordinates, we execute the following optimization formula:

$$Nodes_{initial} = \arg \max_{a,b,i,j} \sum H(a, b, r) \quad (\text{eq.22})$$

3.2) Spatial Relationship Modelling: Distance Matrix Calculation

Prior to spatial pruning, the model constructs a comprehensive distance matrix (D). This matrix serves as the cost-lookup table for the subsequent Constraint Satisfaction Problem (CSP), quantifying the **Pairwise Euclidean Distance** between all i and j detected points.

The distance between any two points P_i and P_j in R^2 is calculated using the Norm-2 formula:

$$d_{ij} = \| P_i - P_j \|_2 = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2} \quad (\text{eq.23})$$

Matrix Properties

Matrix D is characterized by the following mathematical properties:

Symmetry: $d_{ij} = d_{ji}$, confirming the reciprocal nature of distance between nodes.

Zero Diagonal: $d_{ii} = 0$, representing the distance of a node to its own coordinate.

3.2) Spatial Optimization: Greedy Algorithm and Constraint Satisfaction

To refine the network topology, we apply a "**Spatial Filter**" to prune the initial nodes down to an optimal set. This is modelled as a **Constraint Satisfaction Problem (CSP)** using a Greedy Algorithm. The algorithm iteratively selects points from the total set while discarding any candidates that violate the spatial threshold τ . This selection logic is governed by the following function:

$$S_{final} = \{P_i \in S_{total} \mid \forall P_j \in S_{final}, d(P_i, P_j) \geq \tau\} \quad (\text{eq.24})$$

Constraint Parameters

To mitigate near-field interference and optimize the signal-to-interference ratio, we enforce a hard distance threshold:
 $\tau = 50$

Objective Function

Goal: Maximize Coverage

Subject to: $d(P_i, P_j) > 50$

4. Phased Antenna Array

4.1) Phased Array Configuration and Beam Steering

To enhance signal directivity towards specific faculty zones, a Phased Array system is modelled. The steering capability is achieved by applying a progressive phase shift to each of the N antenna elements. For a desired steering angle θ , the phase shift ϕ_n for the n^{th} element is calculated as:

$$\phi_n = -(n - 1) \cdot k_0 \cdot d \cdot \sin(\theta) \quad (\text{eq.25})$$

Where $k_0 = \frac{2\pi}{\lambda}$ is the free-space wave number, and d is the inter-element spacing (**set to** $\frac{\lambda}{2}$). The resulting complex amplitude A_n for each source in the array is defined as:

$$A_n = A_{source} \cdot e^{j\phi_n} \quad (\text{eq.26})$$

4.2) Wave Propagation Modelling

The spatial field distribution for the phased array is modelled using the inhomogeneous Helmholtz equation. This formulation integrates the propagation principles and atmospheric attenuation (α) established in **Section 1.1**. These physical properties are embedded into the complex wave number $k(x, y)$, which incorporates both the refractive index n_{eff} and the medium's attenuation (α):

$$k = k_0 \cdot n_{eff} - j\alpha \quad (\text{eq.27})$$

By incorporating these parameters into the complex wave number (k), the model simulates the real-world signal behavior across the faculty's spatial grid using the **Finite Difference Method (FDM)** as in **section 1.2**

4.3) Coverage Synthesis and Heatmap Generation

Since the array performs an angular sweep to cover the entire sector, the final coverage map is synthesized using a **Max-Hold algorithm**. This process extracts the maximum signal intensity at each spatial coordinate (x, y) from all simulated steering angles θ :

$$E_{final}(x, y) = \max_{\theta} |E(x, y, \theta)| \quad (\text{eq.28})$$
